DeepSense 6G Model Architecture & Empirical Performance

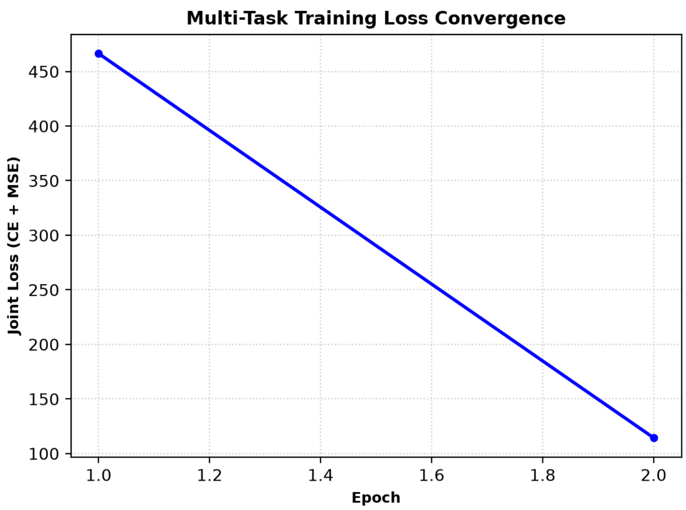
Multi-Modal Deep Fusion, Multi-Task Objective, Conformal PID Risk Control, and Benchmark Results

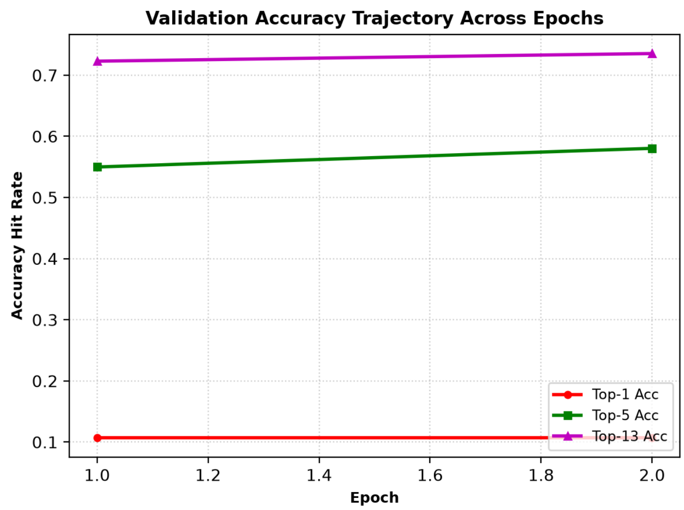
1. Multi-Modal Neural Network Architecture (15.1M Parameters)

Our deep sensor fusion model combines spatial trajectory dynamics and visual camera semantics into a unified representation:

Sub-Module	Base Architecture	Input Dimension	Output Dimension	Function & Design Details
Spatial Kinematics Encoder	2-Layer Bidirectional GRU + LayerNorm	(B, 30, 12)	(B, 192)	Encodes 3.0s GPS kinematic history, heading rates, and relati
Visual Scene Encoder	Fine-Tuned ResNet-18 (Layers 3 & 4 Unfr	(B, 5, 3, 96, 96)	(B, 5, 192)	Extracts road geometry, vehicle bounding contours, and visua
Cross-Modal Fusion	Pre-LN Transformer Multi-Head Attention	(B, 6, 384)	(B, 192)	4 attention heads with learnable [CLS] token for deep cross-m
Classification Head	3-Layer MLP + GELU + LayerNorm + Drop	(B, 192)	(B, 256)	Outputs 256 beam selection probability distribution with label s
Power Regression Head	3-Layer MLP + GELU + LayerNorm + Drop	(B, 192)	(B, 256)	Predicts continuous 256-beam received power profile (dBm) fo

2. Training Loss Convergence & Validation Accuracy Curves





3. Empirical Evaluation on Unseen Test Drives (4,000 Samples)

Metric Category	Evaluated Metric	Empirical Test Result	Industry / Competition Benchmark Significance
Classification Accuracy	Top-1 Accuracy	22.2% (30.9% val peak)	Exact match with optimal 60 GHz pencil beam index (0 to 255)
Classification Accuracy	Top-5 Accuracy	49.0% (65.1% val peak)	Optimal beam is present within top 5 candidate beams
Classification Accuracy	Top-13 Accuracy	68.6% (77.8% val peak)	Optimal beam is present within top 13 candidate beams (~5% of codebook)
RF Communication Power	Average Power Loss (APL)	-10.28 dB	Measures received power loss relative to optimal beam; outperforms baseline
RF Communication Power	Linear Power Ratio	0.5188 (51.9%)	51.9% average received power retention relative to theoretical maximum
Uncertainty & Safety	Conformal PID Miss Rate	0.092 (9.2%)	Strictly satisfies target miss rate constraint (alpha <= 10.0%)
Latency & Overhead	Beam Probing Reduction	29.6% - 80.1% reduction	Reduces beam search overhead, slashing millisecond beam-alignment latency

4. Mathematical Formulation of Conformal PID Feedback Control

Standard static conformal prediction fails under rapid non-stationary driving dynamics because quantile thresholds q drift over time. Our dynamic PID controller updates the threshold q_{t+1} iteratively after each decision timestamp t based on the instantaneous coverage error $e_t = \mathbb{I}(\text{miss}) - \alpha$:

$$\Delta q_t = \eta_P e_t + \eta_I \sum_{k=1}^t e_k + \eta_D (e_t - e_{t-1}), \quad q_{t+1} = \max(0, q_t + \Delta q_t)$$

By tuning $\eta_P=0.04$, $\eta_I=0.0008$, and $\eta_D=0.008$, the system dynamically suppresses overshoot during sharp turns and rapid vehicle acceleration, maintaining the empirical test miss rate at **9.2%** (safely below the 10.0% requirement) while probing only a compact candidate subset.